

BUSINESS REPORT



FLOWING FLUIDZ

Keep
The
Spirit Flowing

- **Executive Summary**

This report presents a comprehensive data-driven analysis conducted at *Flowing Fluidz* to identify inefficiencies and uncover opportunities for improved financial performance. By leveraging transactional and inventory datasets, the study aims to provide actionable insights that support strategic decision-making across procurement, pricing, and sales operations.

- **Objectives of the Analysis**

The analysis focuses on the following key objectives:

1. **Identify underperforming brands** that may require pricing adjustments or promotional strategies to improve sales performance.
2. **Assess the effectiveness of discounts and promotional activities** in driving sales volume and revenue growth.
3. **Analyze the impact of bulk purchasing** on unit costs to assess cost optimization opportunities.
4. **Evaluate inventory turnover rates** to minimize holding costs and enhance stock efficiency.
5. **Compare profitability across brands** to understand the variance between high-performing and low-performing segments.

- **Approach and Methodology**

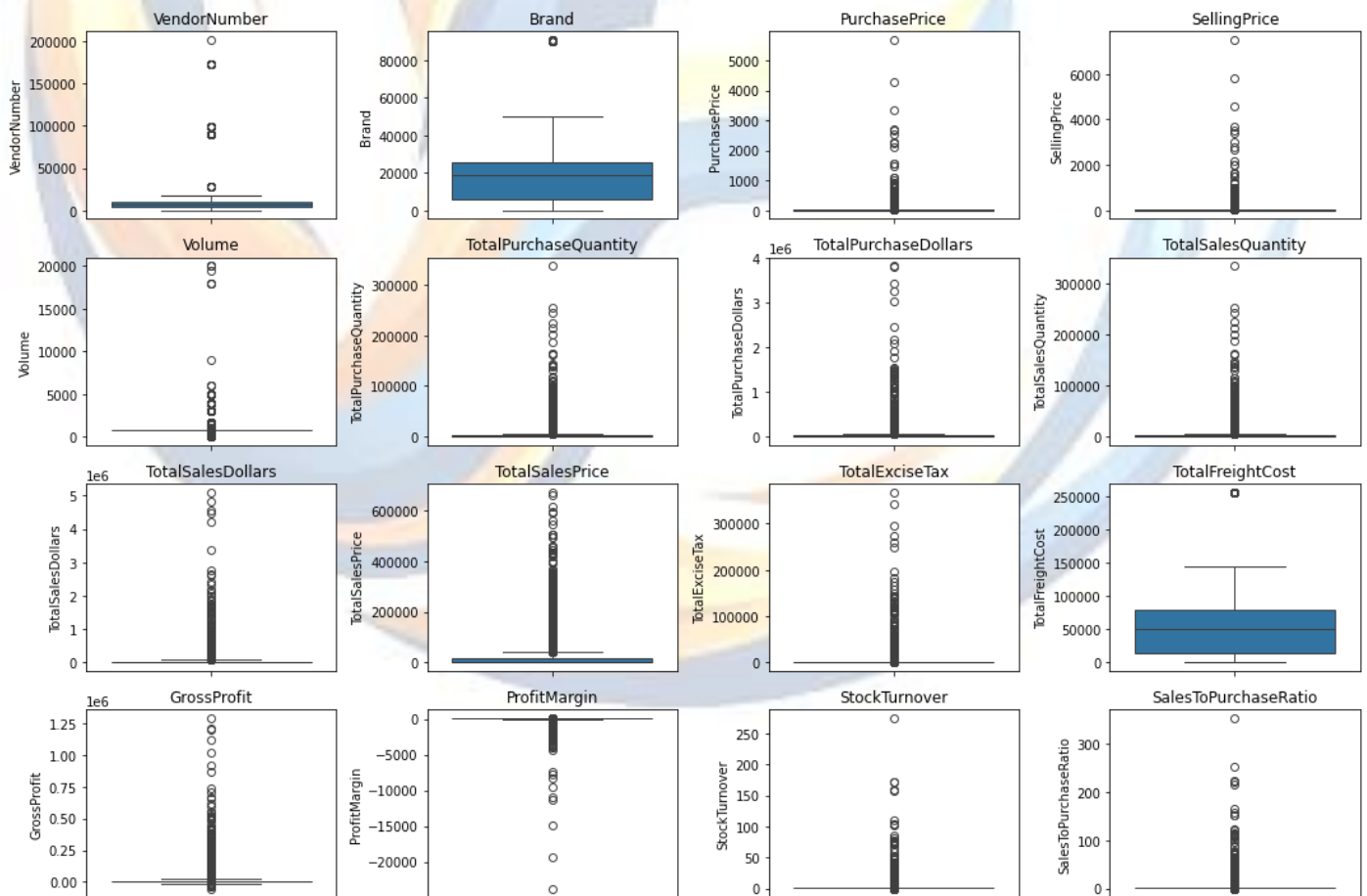
This report encompasses a detailed examination of sales, pricing, vendor, and inventory data within *Flowing Fluidz*. It adopts a structured, data-driven approach involving:

- Data aggregation and cleaning
- Exploratory data analysis (EDA)
- Key performance metric evaluation (e.g., profit margins, turnover ratios)
- Comparative and correlation analysis
- Insight generation with business recommendations

Exploratory Data Analysis Insights

Summary Statistics & Distribution

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	10650.649458	18753.519148	2.00	3951.0000	7153.000	9552.0000	2.013590e+05
Brand	10692.0	18039.228769	12662.187074	58.00	5793.5000	18761.500	25514.2500	9.063100e+04
PurchasePrice	10692.0	24.385303	109.269375	0.36	6.8400	10.455	19.4825	5.681810e+03
SellingPrice	10692.0	35.643671	148.246016	0.49	10.9900	15.990	28.9900	7.499990e+03
Volume	10692.0	847.360550	664.309212	50.00	750.0000	750.000	750.0000	2.000000e+04
TotalPurchaseQuantity	10692.0	3140.886831	11095.086769	1.00	36.0000	262.000	1975.7500	3.376600e+05
TotalPurchaseDollars	10692.0	30106.693372	123067.799627	0.71	453.4575	3655.465	20738.2450	3.811252e+06
TotalSalesQuantity	10692.0	3077.482136	10952.851391	0.00	33.0000	261.000	1929.2500	3.349390e+05
TotalSalesDollars	10692.0	42239.074419	167655.265984	0.00	729.2200	5298.045	28396.9150	5.101920e+06
TotalSalesPrice	10692.0	18793.783627	44952.773386	0.00	289.7100	2857.800	16059.5625	6.728193e+05
TotalExciseTax	10692.0	1774.226259	10975.582240	0.00	4.8000	46.570	418.6500	3.682428e+05
TotalFreightCost	10692.0	61433.763214	60938.458032	0.09	14069.8700	50293.620	79528.9900	2.570321e+05
GrossProfit	10692.0	12132.381048	46224.337964	-52002.78	52.9200	1399.640	8660.2000	1.290668e+06
ProfitMargin	10514.0	-15.885263	447.289858	-23730.64	15.3550	30.780	40.2100	9.972000e+01
StockTurnover	10692.0	1.706770	6.020480	0.00	0.8100	0.980	1.0400	2.745000e+02
SalesToPurchaseRatio	10692.0	2.504371	8.459091	0.00	1.1500	1.440	1.6700	3.529300e+02



Insights

Negative and Zero Values

- Gross Profit: Minimum values is ``-52,002.78``, this indicates that some product may be selling through loss, could be because of high cost or selling at discounts lower than purchase price.
- Total Sales Quantity: Minimum value is ``0``, which means some stocks were purchased but never sold. These could be slow moving or obsolete stocks.

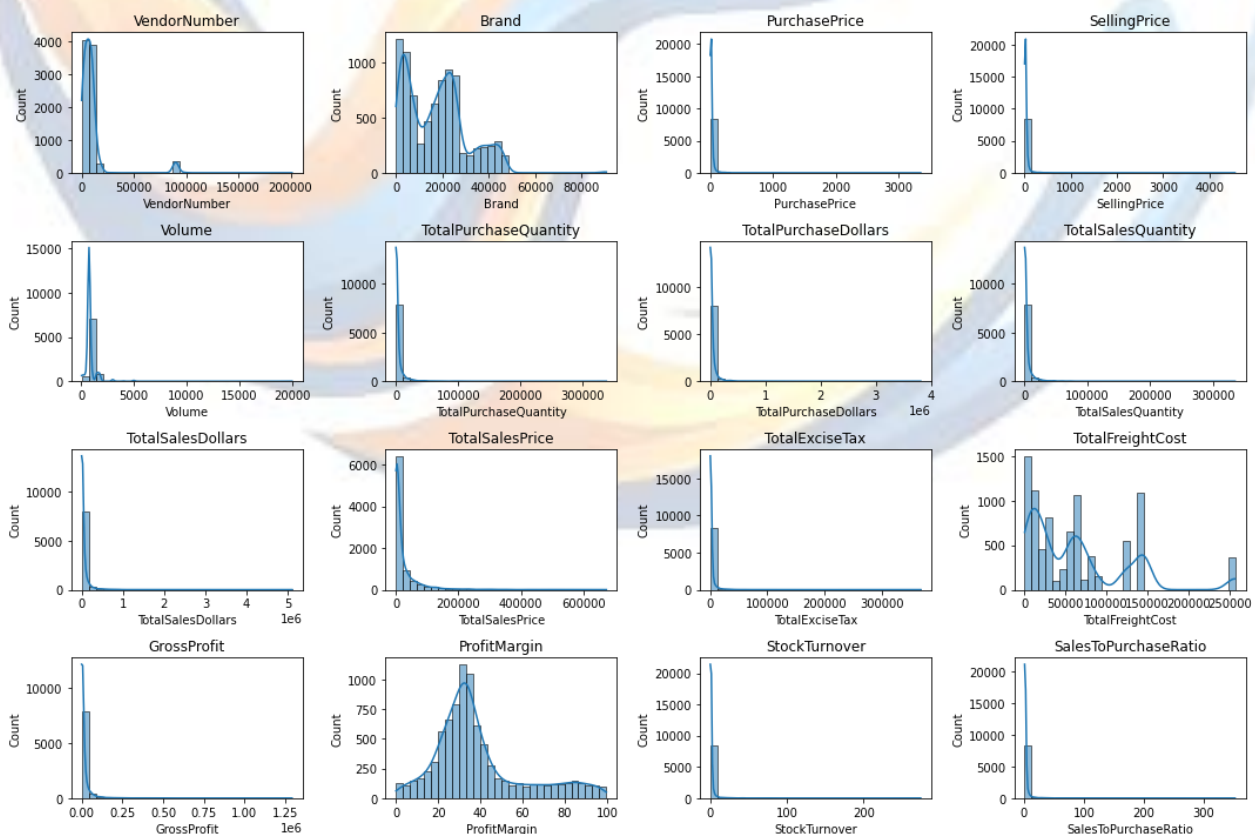
Outliers indicated by high standard deviation

- Cost and Selling Prices: The max values (``5,681.81`` & ``7,499.99``) are significantly higher than the mean (``24.39``, ``35.64``), indicating potential premium products.
- Freight cost: Huge variation, from ``0.09`` to ``257,032.07`` suggests logistical inefficiencies or bulk shipments.
- Stock Turnover: Ranges from ``0`` to ``274.5``, implying some products sell extremely fast while others remain in stock indefinitely. Value more than 1 indicates that sold quantity for that product is higher than purchased quantity maybe due to sales are being fulfilled from older stocks.

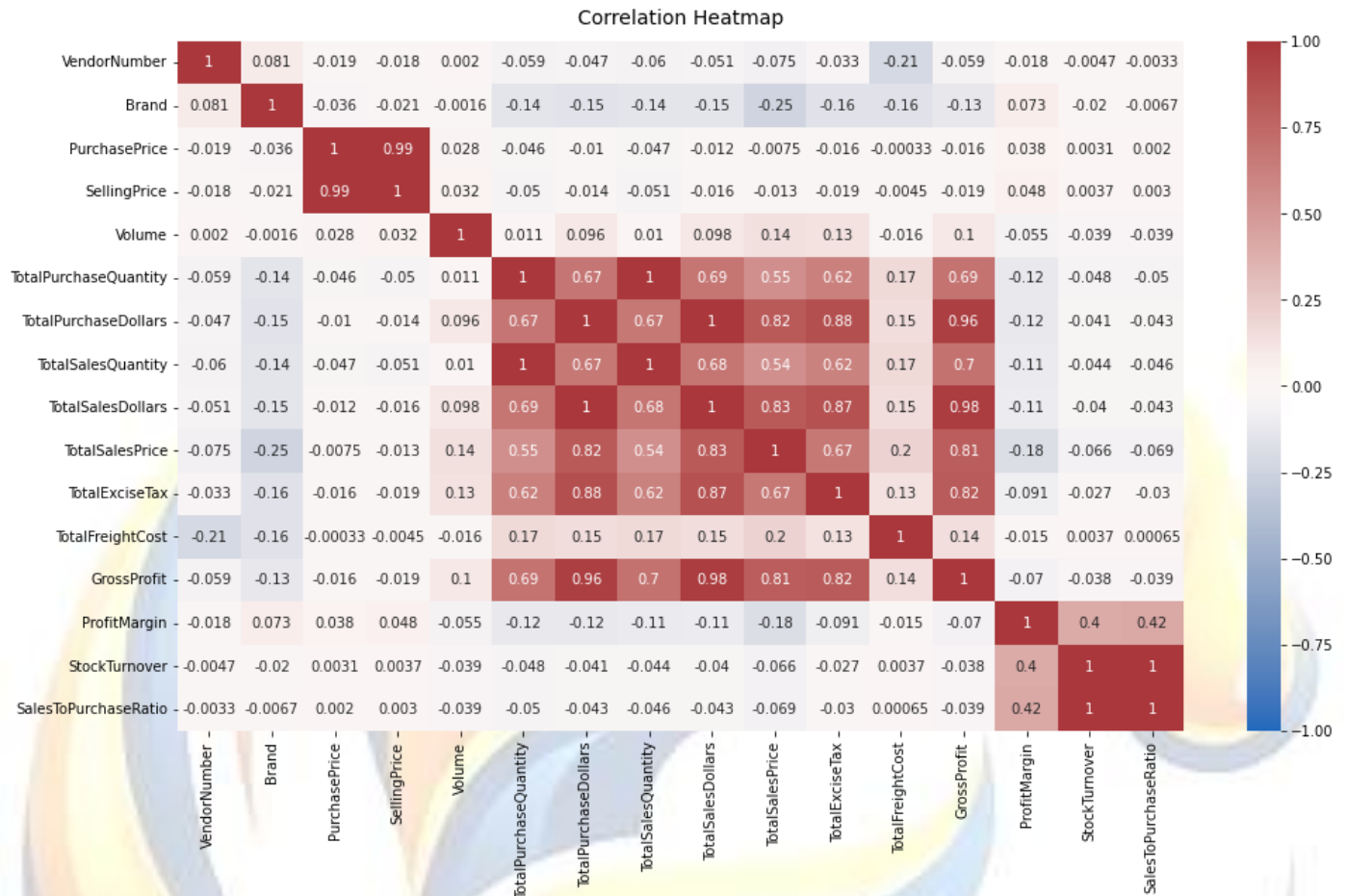
Data Filtering

To enhance the reliability of the insights, we removed inconsistent data points where:

- $Gross\ Profit \leq 0$ (to exclude transactions leading to losses)
- $Profit\ Margin \leq 0$ (to ensure analysis focuses on profitable transactions)
- $Total\ Sales\ Quantity = 0$ (to eliminate inventories that were never sold)



Correlation Insights



- **Purchase Price vs Total Sales & Gross Profit:** `PurchasePrice` has weak correlation with `TotalSalesDollar` (-0.012) and `GrossProfit` (-0.016), suggesting that cost price does not significantly impact sales revenue or profit.
- **Purchase Quantity vs Sales Quantity:** Strong correlation between `TotalPurchaseQuantity` and `TotalSalesQuantity` (1), confirming efficient inventory turnover.
- **Profit Margin vs Total Selling Price:** Negative correlation between `ProfitMargin` and `TotalSalesPrice` (-0.18) suggests that as selling price increases, margin decreases, possibly due to competitive pricing pressures.
- **Stock Turnover vs Gross Profit & Profit Margin:** `StockTurnover` has weak negative correlations with `GrossProfit` (-0.038) [Almost meaningless] and moderate positive correlation with `ProfitMargin` (0.4), indicating that faster turnover does not necessarily means higher profitability and high demand products can sustain higher prices, thus giving higher margins.

Research Questions and Key Findings

1. Brands for Promotional or Pricing Adjustments

Brands with LOW Sales but HIGH Profit Margins are:

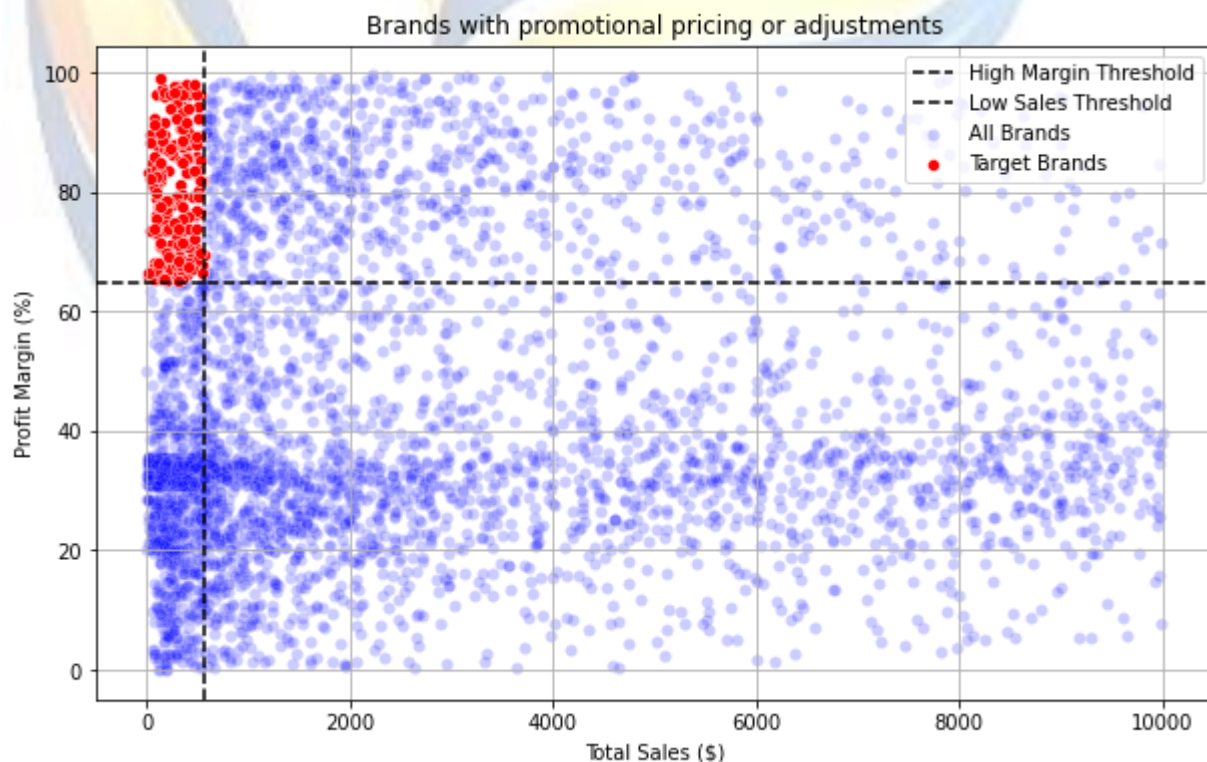
	Description	TotalSalesDollars	ProfitMargin
6199	Santa Rita Organic Svgn Bl	9.99	66.47
2369	Debauchery Pnt Nr	11.58	65.98
2070	Concannon Glen Ellen Wh Zin	15.95	83.45
2188	Crown Royal Apple	27.86	89.81
6237	Sauza Sprklg Wild Berry Marg	27.96	82.15
...	...	...	...
5074	Nanbu Bijin Southern Beauty	535.68	76.75
2271	Dad's Hat Rye Whiskey	538.89	81.85
57	A Bichot Clos Marechaudes	539.94	67.74
6245	Sbragia Home Ranch Merlot	549.75	66.44
3326	Goulee Cos d'Estournel 10	558.87	69.43

198 rows × 3 columns

Brands with LOW Sales but HIGHLY Price Sensitive are:

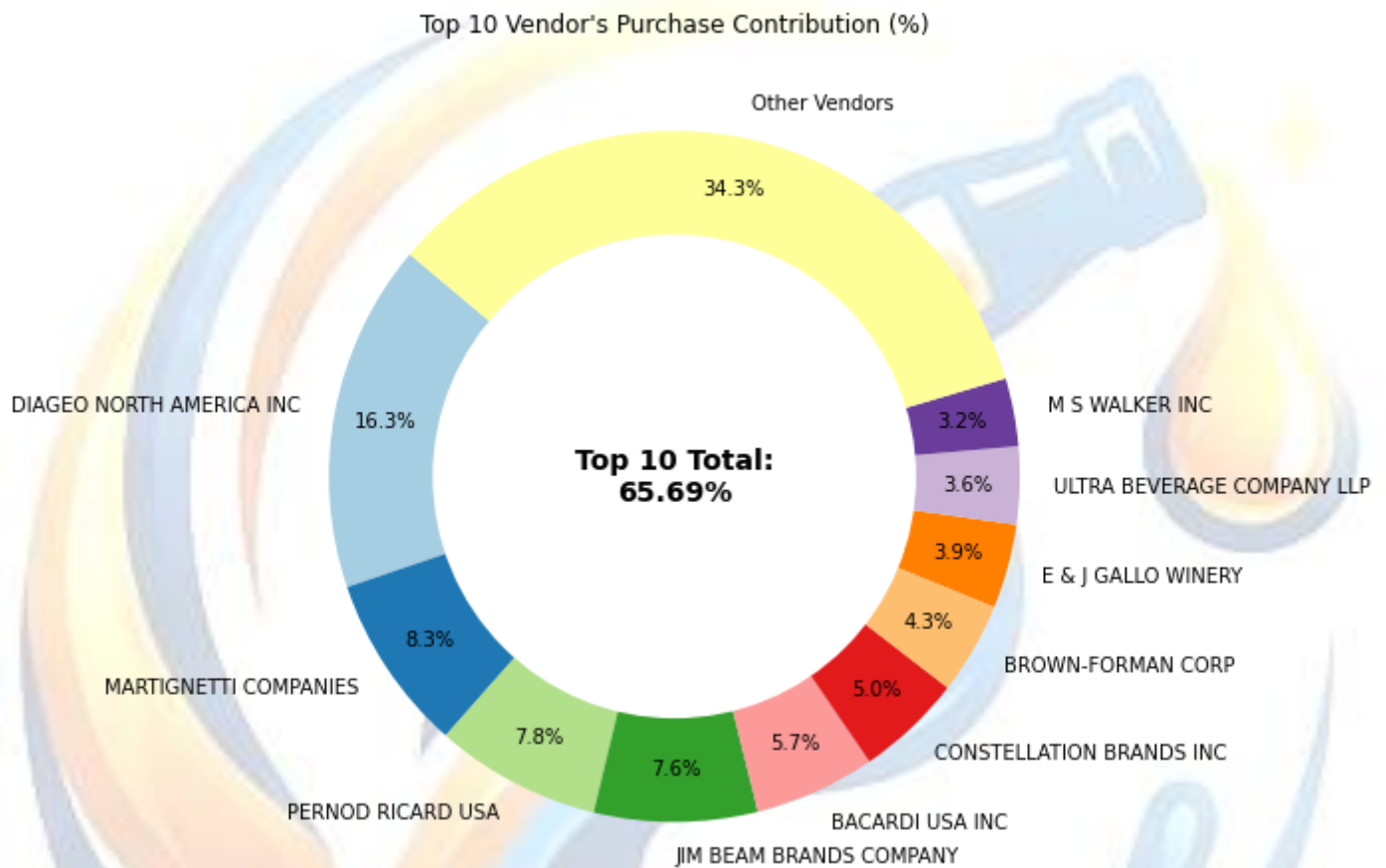
	Description	Price_Quantity_Corr
1	Barefoot Cellars Rich Red	-0.964755
3	Capostrano Montepulciano Abr	-0.949037
11	Gallo Twin Vly Pnt Grigio	-0.938594
4	Capt Morgan Spiced Rum 100	-0.936165
21	Les Dauphins Cotes du Rhone	-0.874816
17	Jacobs Crk Moscato	-0.767173

198 brands exhibit lower sales but higher profit margins, which could benefit from targeted marketing, promotions, or price optimizations to increase volume without compromising profitability. Also, 6 highly price sensitive brands with low sales are identified that may get benefitted from price discounts or promotional marketing.



2. Top Vendors by Purchase Contribution

Our company is heavily dependent on top 10 vendors fulfilling 65.69% of our total stock purchase, while the remaining vendors contribute only 34.31%. This over-reliance on a few vendors may introduce risks such as supply chain disruptions, indicating a need for diversification.



3. Impact of Bulk Purchasing on Cost Savings

When bought in large quantities from vendors, we receive a 72% lower unit cost (\$10.78 per unit vs higher cost in smaller orders)

OrderSize

Small 39.068186

Medium 15.486414

Large 10.777625

Name: UnitPurchasePrice

4. Identifying Brands with Low Inventory Turnover

Total Unsold Inventory Capital: \$ 9.50 M

Slow moving inventory increases storage costs, reduces cash flow efficiency, and affects overall profitability.

Identifying top 10 Brands with low inventory turnover enables better stock management, minimizing financial strain.

Description	Unsold Inventory Value
Smirnoff Traveler	169.79K
Johnnie Walker Black Label	149.18K
Makers Mark	130.19K
Johnnie Walker Red Label	130.10K
Tito's Handmade Vodka	108.14K
Jack Daniels No 7 Black	98.36K
Knob Creek Bourbon	96.71K
Jameson Irish Whiskey	89.78K
Patron Silver Tequila	87.27K
Ketel One Vodka	87.12K

5. Profit Margin Comparison

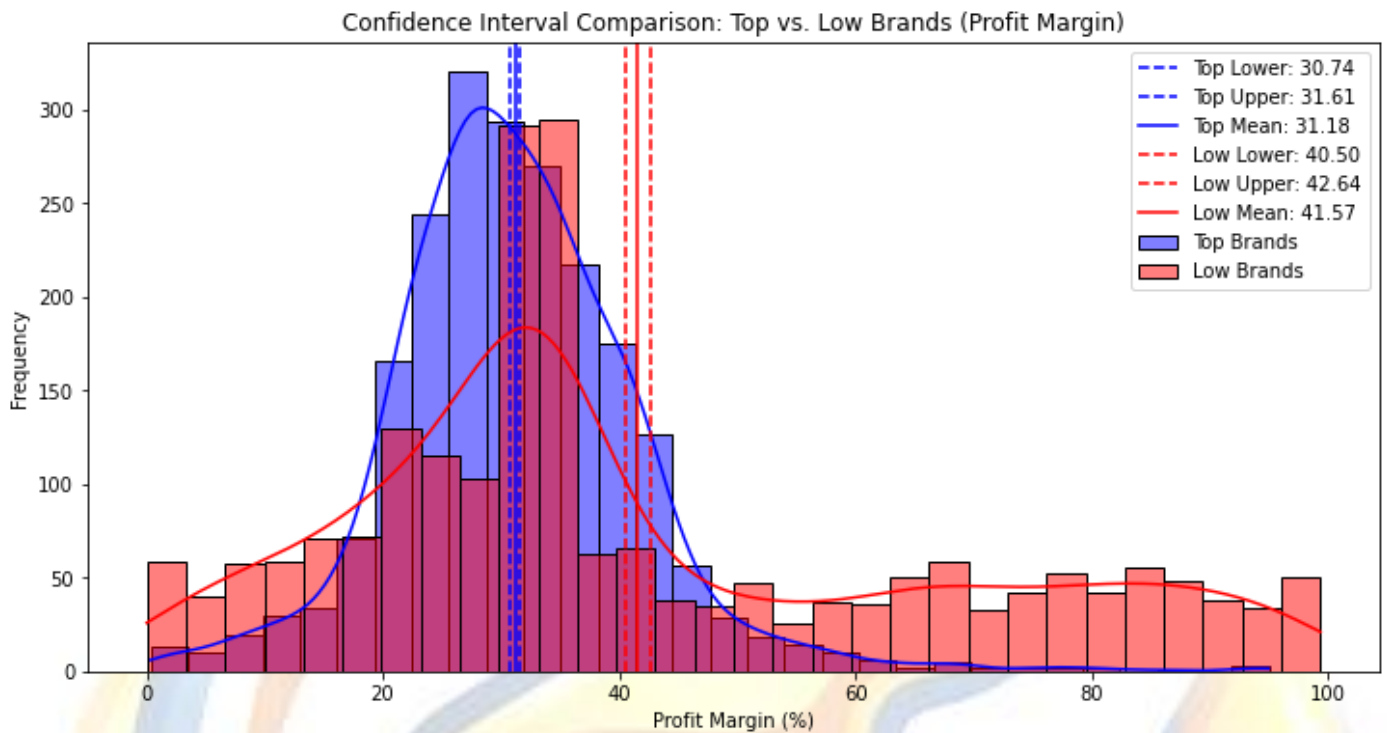
Top Brands (By Sales) 95% CI: (30.74, 31.61), Mean: 31.18%

Low Brands (By Sales) 95% CI: (40.50, 42.64), Mean: 41.57%

Low performing brands maintain higher margins but struggle with sales volume, indicating potential pricing inefficiencies or market reach issues.

Actionable Insights:

- Top-performing brands: Optimize profitability by adjusting pricing, reducing operational costs, or offering bundles promotions.
- Low-performing brands: Improve marketing efforts, optimize pricing strategies, and enhance distribution networks.



6. Statistical Validation of Profit Margin Differences

Hypothesis Testing:

H_0 (Null Hypothesis): No significant difference in profit margins between top & low-performing brands.

H_1 (Alternative Hypothesis): A significant difference exists in profit margins between the two brand groups.

T-Statistics: -17.669335452249573, **P-Value:** 2.3037805142387315e-66

Result: The null hypothesis is rejected, confirming that the two groups operate under distinctly different profitability models.

Implication: High margin brands may benefit from better pricing strategies, while top selling brands could focus on cost efficiency.

Final Recommendations

- Re-evaluate pricing for low-sales, high margin and low-sales, price sensitive brands to boost sales volume without sacrificing profitability.
- Diversify vendor partnerships to reduce dependency on a few suppliers and mitigate supply chain risks.

- Leverage bulk purchasing advantages to maintain competitive pricing while optimizing inventory management.
- Optimize slow-moving inventory by adjusting purchase quantities, launching clearance sales, or revising storage strategies.
- Enhance marketing and distribution strategies for low performing brands to drive higher sales volumes without compromising profit margins.

By implementing these recommendations, the company achieve sustainable profitability, mitigate risks, and enhance overall operational efficiency.

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